20261124 Quiz

for (MATH301) Modern Algebra I

Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology

Exercise 7.3.34. Let I and J be ideals of a ring R.

(a) Prove that $I + J$ is the smallest ideal containing both I and J.

(b) Prove that IJ is an ideal contained in $I \cap J$.

(c) Give an example where $IJ \neq I \cap J$.

(d) Prove that if R is commutative and $I + J = R$, then $IJ = I \cap J$.

Solution.

(a) The set $I + J = \{a + b \mid a \in I, b \in J\}$ is an additive subgroup and is closed under left and right multiplication by elements of R. It contains I and J since 0 lies in both ideals. Any ideal containing both must contain every $a + b$, proving minimality.

(b) By definition, IJ consists of finite sums $\sum_k a_k b_k$ with $a_k \in I$, $b_k \in J$. It is closed under addition and additive inverses. Also $r(a_k b_k) = (ra_k)b_k$ and $(a_k b_k)r = a_k(b_k r)$ lie in IJ, so it is an ideal. Each $a_k b_k$ belongs to both I and J, hence $IJ \subseteq I \cap J$.

(c) In $R = \mathbb{Z}$, take $I = J = 2\mathbb{Z}$. Then $IJ = 4\mathbb{Z} \neq 2\mathbb{Z} = I \cap J$.

(d) Suppose $1 = a + b$ with $a \in I$, $b \in J$. If $x \in I \cap J$, then $x = xa + xb \in IJ$, since R is commutative. Together with (b), this gives $I \cap J = IJ$.

□

Exercise 7.3.36. Let I be the ideal of all polynomials in $\mathbb{Z}[x]$ with zero constant term. Show that

$$I^n = \left\{ \sum_{i=n}^{n+m} a_i x^i \;\middle|\; a_i \in \mathbb{Z},\ m \geq 0 \right\}$$

(with the zero polynomial also included), i.e., the polynomials whose first nonzero term has degree at least n.

Solution. Since $I = (x)$, the product of n elements of I is divisible by x^n, so $I^n \subseteq (x^n)$. Conversely, x^n is a product of n copies of $x \in I$, and I^n is an ideal; therefore $(x^n) \subseteq I^n$. Thus $I^n = (x^n)$, exactly the polynomials with all coefficients of degrees below n equal to zero.

□

Exercise 7.3.37. An ideal N is called *nilpotent* if $N^n = (0)$ for some $n \geq 1$. Prove that the ideal $p\mathbb{Z}/p^m\mathbb{Z}$ is nilpotent in the ring $\mathbb{Z}/p^m\mathbb{Z}$.

Solution. Write $N = (\,[p]_{p^m}\,)$. The product of m elements of N is divisible by p^m and hence is zero modulo p^m. Therefore $N^m = (\,[p^m]_{p^m}\,) = (0)$, as required.

□

Exercise 7.4.5. Prove that if M is an ideal of a ring R such that R/M is a field, then M is a maximal ideal. Do not assume R is commutative.

Solution. Since R/M is a nonzero field, $M \neq R$. If J is a (two-sided) ideal of R containing M, then J/M is an ideal of R/M. The only ideals of a field are (0) and the whole field, so $J/M = (0)$ or $J/M = R/M$. Thus $J = M$ or $J = R$. Hence M is maximal, without any commutativity assumption on R.

□

Exercise 7.4.7. Let R be a commutative ring with 1. Prove that the principal ideal (x) of $R[x]$ is a prime ideal if and only if R is an integral domain, and is a maximal ideal if and only if R is a field.

Solution. Evaluation at 0 defines a surjective ring homomorphism

$$\varepsilon_0 : R[x] \to R, \qquad f(x) \mapsto f(0),$$

whose kernel is (x). Hence $R[x]/(x) \cong R$. By the quotient criteria for prime and maximal ideals in a commutative ring, (x) is prime exactly when R is an integral domain and maximal exactly when R is a field.

□

Exercise 7.4.8. Let R be an integral domain. Prove that $(a) = (b)$ for $a, b \in R$ if and only if $a = ub$ for some unit $u \in R^\times$.

Solution. If $a = ub$ with u a unit, then each of a and b is a multiple of the other, so $(a) = (b)$. Conversely, suppose $(a) = (b)$. If $a = 0$, then $b = 0$ and we may take $u = 1$. Otherwise, $a = rb$ and $b = sa$ for some $r, s \in R$. Thus $a = rsa$, and cancellation in the integral domain yields $rs = 1$. Therefore $r \in R^\times$, and $a = rb$.

□

Exercise 7.4.10. Assume R is commutative. Prove that if P is a prime ideal of R and P contains no zero divisors, then R is an integral domain.

Solution. Let $ab = 0$ in R. Since $0 \in P$ and P is prime, either $a \in P$ or $b \in P$. If both a and b were nonzero, each would be a zero divisor, contradicting the hypothesis on P. Therefore $a = 0$ or $b = 0$. As R has $1 \neq 0$ by the standing assumption of §7.4, it is an integral domain.

□

Exercise 7.4.18. Let R be an integral domain and let $R[[x]]$ be the formal power series ring. Prove that the principal ideal (x) is prime. Prove that (x) is maximal if and only if R is a field (cf. Exercise 7.2.3).

Solution. Taking the constant coefficient defines a surjective ring homomorphism

$$\varepsilon : R[[x]] \to R, \qquad \sum_{n \geq 0} a_n x^n \mapsto a_0.$$

Its kernel comprises the series with zero constant coefficient, precisely (x). Thus $R[[x]]/(x) \cong R$. Because R is an integral domain, (x) is prime; by the maximal-ideal criterion, it is maximal exactly when R is a field.

□

Exercise 7.4.19. Let R be a finite commutative ring with identity. Prove that every prime ideal of R is maximal.

Solution. For a prime ideal P, the quotient R/P is a finite integral domain. If $0 \neq \overline{a} \in R/P$, multiplication by $\overline{a}$ is injective and hence, by finiteness, surjective. Some $\overline{b}$ therefore satisfies $\overline{a}\,\overline{b} = 1$. Thus R/P is a field, so P is maximal.

□

Exercise 7.4.21. Prove that a finite ring with identity $1 \neq 0$ and no zero divisors is a field. You may quote Wedderburn's Theorem.

Solution. For each $0 \neq a \in R$, both maps $r \mapsto ar$ and $r \mapsto ra$ are injective, since R has no zero divisors. Because R is finite, both are surjective. Hence there exist $b, c \in R$ such that $ab = 1 = ca$. Then $c = c(ab) = (ca)b = b$, so a is invertible. Thus R is a finite division ring. Wedderburn's Little Theorem states that every finite division ring is commutative; consequently R is a field.

□

Exercise 7.4.25. Assume R is commutative and that for each $a \in R$ there is an integer $n > 1$ (depending on a) such that $a^n = a$. Prove that every prime ideal of R is maximal.

Solution. Let P be a prime ideal. Then R/P is an integral domain. For any nonzero $\overline{a} \in R/P$, choose $n > 1$ such that $a^n = a$. We obtain

$$\overline{a}(\overline{a}^{\,n-1} - 1) = 0.$$

Since $\overline{a} \neq 0$ and the quotient has no zero divisors, $\overline{a}^{\,n-1} = 1$. Thus $\overline{a}$ is a unit (with inverse $\overline{a}^{\,n-2}$). Every nonzero element of R/P is a unit, so R/P is a field and P is maximal.

□

Exercise 7.4.33. Let $R = C([0,1], \mathbb{R})$ be the ring of continuous real-valued functions on $[0,1]$. For $c \in [0,1]$, let $M_c = \{f \in R \mid f(c) = 0\}$, a maximal ideal of R.

(a) Prove that every maximal ideal M of R is M_c for some $c \in [0,1]$.

(b) Prove that $M_b \neq M_c$ if $b \neq c$.

(c) Prove that M_c is not the principal ideal generated by $x - c$.

(d) Prove that M_c is not finitely generated.

Solution.

(a) Suppose that no point of $[0,1]$ is a common zero of all functions in M. For each $c \in [0,1]$ choose $f_c \in M$ with $f_c(c) \neq 0$. The open sets where f_c does not vanish cover $[0,1]$. Compactness gives a finite subcover from $f_1, \ldots, f_m \in M$. Then $h = f_1^2 + \cdots + f_m^2 \in M$ is strictly positive everywhere, so $1/h \in R$ and $1 = h(1/h) \in M$, a contradiction. Hence some c satisfies $f(c) = 0$ for every $f \in M$, i.e., $M \subseteq M_c$. Maximality gives $M = M_c$.

(b) The function $f(x) = x - b$ lies in M_b but not in M_c whenever $b \neq c$.

(c) Let $t(x) = x - c$ and $g(x) = \sqrt{|x - c|}$. Both vanish at c, so $g \in M_c$. If $g = th$ for a continuous h, then for $x \neq c$,

$$|h(x)| = \frac{1}{\sqrt{|x - c|}},$$

which is unbounded as $x \to c$ within $[0,1]$, a contradiction. Thus $(t) \neq M_c$.

(d) Suppose $M_c = (f_1, \ldots, f_m)$. Put $h = \sum_{i=1}^m |f_i|$. Then $h \geq 0$ is continuous, $h(c) = 0$, and h is nonzero at points arbitrarily close to c: otherwise all f_i vanish on a neighborhood of c, but $x - c \in M_c$ does not. As $\sqrt{h} \in M_c$, there exist continuous g_i with $\sqrt{h} = \sum_i f_i g_i$. By continuity, the g_i are bounded, say $|g_i| \leq B$ on $[0, 1]$ for some $B > 0$. Hence

$$\sqrt{h(x)} \leq \sum_i |f_i(x)||g_i(x)| \leq Bh(x).$$

At points with $h(x) > 0$ approaching c, this gives $1 \leq B\sqrt{h(x)} \to 0$, a contradiction. Therefore M_c is not finitely generated.

□

Exercise 7.4.37. A commutative ring R is called a *local ring* if it has a unique maximal ideal. Prove that if R is local with maximal ideal M, then every element of $R \setminus M$ is a unit. Conversely, prove that if R is a commutative ring with 1 and the set of nonunits is an ideal M, then R is local with unique maximal ideal M.

Solution. Assume M is the unique maximal ideal. If $a \notin M$ were a nonunit, the principal ideal (a) would be proper and hence contained in some maximal ideal, necessarily M. This would imply $a \in M$, a contradiction. Thus every $a \notin M$ is a unit.

Conversely, if the nonunits form an ideal M, then M is proper since 1 is a unit. Every proper ideal contains no unit and is therefore contained in M. Hence M is maximal and is the unique maximal ideal.

□

Exercise 7.4.38. Prove that the ring of rational numbers with odd denominators is a local ring whose unique maximal ideal is the principal ideal generated by 2.

Solution. Let

$$R = \left\{ \frac{a}{b} \in \mathbb{Q} \;\middle|\; a, b \in \mathbb{Z},\ b \text{ odd} \right\}.$$

A nonzero fraction $a/b \in R$ is a unit precisely when its numerator a is odd: its inverse b/a belongs to R exactly in that case (after reducing the fraction, if necessary). Thus the nonunits are exactly the fractions with even numerator, namely $2R$. This is a proper ideal, so Exercise 7.4.37 shows that R is local with unique maximal ideal $(2) = 2R$.

□